

Advanced Hunting & Proactive Threat Hunting

Hypothesis-Driven KQL Hunting Across Endpoint, Registry, Network & Authentication Telemetry

Status	Environment	Hunting Window	Analyst
Completed	Defender XDR / Defender for Endpoint (SOC lab)	Last 30 days	Edwin — SOC Analyst (L1)

Core Competency	Assets	Primary Tables	Detection Rules Created
Hypothesis-driven KQL hunting	dc01.corp.local, win11-client01.corp.local	DeviceProcessEvents, DeviceRegistryEvents, DeviceNetworkEvents, DeviceLogonEvents	None — hunting only

1. Executive Summary

This project demonstrates a practical proactive threat-hunting workflow using Microsoft Defender XDR Advanced Hunting and Kusto Query Language (KQL). Unlike an alert-driven investigation, the objective was to formulate hunting hypotheses, query available telemetry, identify unusual or low-frequency activity, and determine whether the resulting activity warranted further investigation — without waiting for Defender to raise an alert first.

The portal was consistently operated using the Last 30 days hunting window across four primary tables: DeviceProcessEvents, DeviceRegistryEvents, DeviceNetworkEvents, and DeviceLogonEvents. No detection rules were created from any query — this project is scoped to hunting, not detection engineering.

Telemetry / Hunt	Observed Result	Nature of Count
DeviceProcessEvents baseline	11,229	Raw event count
DeviceRegistryEvents baseline	9,376	Raw event count
DeviceNetworkEvents baseline	8,216	Raw event count
Hunt 1 — Rare process candidates	77	Grouped query result
Hunt 2 — User-writable-path execution	91	Raw matching events
Hunt 3 — Low-frequency parent/child relationships	17	Grouped query result
Hunt 4 — One-device remote-IP groupings	605	Grouped query result
Hunt 5 — Authentication pattern groupings	1,713	Grouped query result — not a raw logon-event count

Scope of the Numbers

The 11,229 / 9,376 / 8,216 figures are raw event counts from unfiltered baseline queries. The hunt results (77, 91, 17, 605, 1,713) are counts of rows returned by grouped or filtered queries — leads to triage, not independent telemetry volumes, and not confirmed threats.

The final assessment was deliberately conservative: the captured hunting evidence did not independently establish a confirmed compromise on either asset.

2. KQL Quick Start — Core Syntax for This Project

Kusto Query Language (KQL) is the query language used by Advanced Hunting to search Defender telemetry. A query starts with a table name, then pipes (|) the result through additional operators — read each | as "take the previous result and pass it to the next step."

Operator	Purpose	Example
where	Filter rows	where FileName == "powershell.exe"
project	Select columns	project Timestamp, DeviceName, FileName
summarize	Aggregate / group data	summarize count() by FileName
dcount()	Approximate distinct count	summarize dcount(DeviceName)
order by	Sort results	order by Timestamp desc
has_any	Match any term in a list (case-insensitive)	where ProcessCommandLine has_any ("EncodedCommand", "WebClient")
in~	Case-insensitive list membership	where FileName in~ ("powershell.exe", "cmd.exe")
isnotempty()	Exclude null/empty values	where isnotempty(RemoteIP)

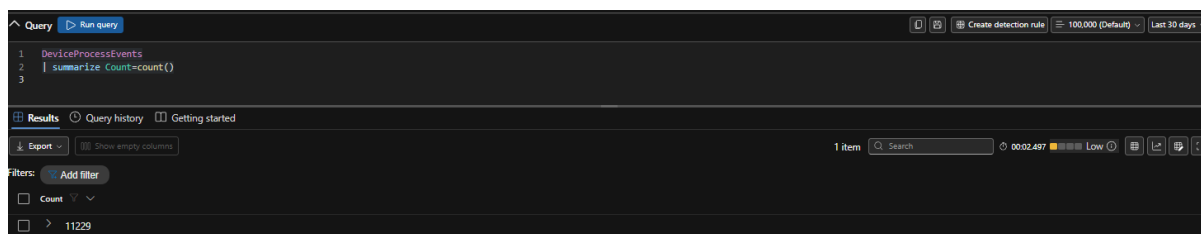
Every hunt in this report follows the same underlying pattern: pick a table, filter or group with where/summarize, sort to surface the most interesting rows with order by, then pivot into the raw event for context. The mental model is table → filter → aggregate → sort → pivot, not memorizing operators in isolation.

3. Environment & Scope

Field	Value
Platform	Microsoft Defender XDR — Advanced Hunting
Assets	dc01.corp.local, win11-client01.corp.local
Hunting period	Last 30 days (fixed for the entire project)
Primary tables	DeviceProcessEvents, DeviceRegistryEvents, DeviceNetworkEvents, DeviceLogonEvents
Detection rules created	None — this is a hunting project, not detection engineering

Telemetry Baseline Validation

Before running any hunt, the three primary tables were validated with a simple count() query to confirm telemetry was actually flowing for the 30-day window — this is environment validation, not portfolio evidence in its own right.



DeviceProcessEvents baseline — 11,229 events.

Count
9376

DeviceRegistryEvents baseline — 9,376 events.

Count
8216

DeviceNetworkEvents baseline — 8,216 events.

4. Hunting Methodology

```
Hunting Hypothesis
|
Select telemetry table
|
Construct KQL
|
Establish baseline
|
Identify anomaly / rarity
|
Select candidate
|
Pivot into context
|
Validate
|
Analyst assessment
```

Core Principle

An anomaly is a lead, not a verdict. A rare process, unusual parent/child relationship, user-writable execution path, uncommon network destination, or low-frequency authentication event can all be entirely legitimate. The analyst must examine additional context before declaring malicious activity.

5. Hunt 1 — Rare Process Execution

Hypothesis: Rarely observed processes may represent unusual software execution requiring further investigation.

Query

```
DeviceProcessEvents
| summarize
    ExecutionCount = count(),
    Devices = dcount(DeviceName),
    Accounts = dcount(AccountName)
    by FileName
| where ExecutionCount <= 3
| order by ExecutionCount asc
```

Result: 77 candidates.

```

1 DeviceProcessEvents
2 | summarize
3   ExecutionCount = count(),
4   Devices = dcount(DeviceName),
5   Accounts = dcount(AccountName)
6   by FileName
7 | where ExecutionCount <= 3
8 | order by ExecutionCount asc

```

FileName	ExecutionCount	Devices	Accounts
> AM_Delta_Patch_1...	1	1	1
> oikFullTrustExecut...	1	1	1
> Microsoft.SharePoi...	1	1	1
> XboxGameBarWid...	1	1	1
> XboxPCAppFT.exe	1	1	1
> cookie_exporter.exe	1	1	1
> regedit.exe	1	1	1
> AM_Delta_Patch_1...	1	1	1
> AM_Delta_Patch_1...	1	1	1
> WaaSMedicAgent...	1	1	1
> Notepad.exe	1	1	1
> AM_Delta_Patch_1...	1	1	1
> AM_Delta_Patch_1...	1	1	1
> AM_Base_Patch1.e...	1	1	1
> VBoxWindowsA...	1	1	1
> VBoxWindowsA...	1	1	1

Evidence — rare process execution results (77 candidates), grouped by FileName, ExecutionCount, Devices, Accounts.

Assessment: The query successfully surfaced low-frequency processes, including AM_Delta_Patch_* update components, Xbox components, cookie_exporter.exe, regedit.exe, WaaSMedicAgent*, Notepad.exe, and VirtualBox-related binaries. Low frequency is not the same as maliciousness — several visible results were consistent with legitimate Windows, security, virtualization, or application activity.

Analyst Conclusion

Hunting leads identified — no confirmed malicious process established.

6. Hunt 2 — Execution from User-Writable Locations

Hypothesis: Executables launched from user-writable directories can warrant investigation because attackers frequently abuse locations such as user profiles, temporary directories, and downloads.

Query

```

DeviceProcessEvents
| where FolderPath has_any (
    @"Users\",
    @"AppData\",
    @"Temp\",
    @"Downloads\",
    @"Public\"
)
| project
    Timestamp, DeviceName, AccountName, FileName, FolderPath,
    ProcessCommandLine, InitiatingProcessFileName,
    InitiatingProcessCommandLine, SHA1
| order by Timestamp desc

```

Result: 91 events.

Query Run query Create detection rule 100,000 (Default) Last 30 days

```

1 DeviceProcessEvents
2 | where FolderPath has_any (
3   @"Users\*",
4   @"AppData\*",
5   @"Temp\*",
6   @"Downloads\*",
7   @"Public\*"
8 )
9 | project Timestamp, DeviceName, AccountName, FileName, FolderPath, ProcessCommandLine, InitiatingProcessFileName, InitiatingProcessCommandLine, SHA1
10 | order by Timestamp desc

```

Results Query history Getting started

Export Show empty columns 91 items Search 0001:215 Low Icons Print Fullscreen

	Timestamp	DeviceName	AccountName	FileName	FolderPath	ProcessCommandLine	InitiatingProcessFi...	InitiatingProcessCo...	SHA1
☐	> Aug 12, 2026 9:00...	win11-client01.corp...	socadmin	OneDriveSetup.exe	C:\Users\Socadmin.COR...	OneDriveSetup.exe /upd...	onedrivesetup.exe	"OneDriveSetup.exe" /u...	42c32e644c4553b07...
☐	> Aug 12, 2026 9:00...	win11-client01.corp...	socadmin	OneDriveSetup.exe	C:\Users\Socadmin.COR...	"OneDriveSetup.exe" /u...	onedrive.exe	"OneDrive.exe" /backgr...	42c32e644c4553b07...
☐	> Aug 12, 2026 8:59...	win11-client01.corp...	socadmin	FileCoAuth.exe	C:\Users\Socadmin.COR...	"FileCoAuth.exe" -Embe...	svchost.exe	svchost.exe -k DcomLau...	68ab9a797cd3e04cb...
☐	> Aug 12, 2026 8:59...	win11-client01.corp...	socadmin	OneDrive.exe	C:\Users\Socadmin.COR...	"OneDrive.exe" /backgr...	explorer.exe	Explorer.EXE	3986df145c2602ec3f...
☐	> Aug 12, 2026 8:57...	dc01.corp.local	system	DismHost.exe	C:\Windows\Temp\CC72...	dismhost.exe (7CD8E4F8...	msense.exe	MsSense.exe	9fd99403b433857f23...
☐	> Aug 11, 2026 11:0...	dc01.corp.local	administrator	DismHost.exe	C:\Users\Administrator\...	dismhost.exe (5F28D668...	cleanmgr.exe	cleanmgr.exe /autoclean...	9fd99403b433857f23...
☐	> Aug 11, 2026 10:5...	dc01.corp.local	system	DismHost.exe	C:\Windows\Temp\95D4...	dismhost.exe (0ADAFBD...	msense.exe	MsSense.exe	9fd99403b433857f23...
☐	> Aug 11, 2026 10:5...	dc01.corp.local	system	DismHost.exe	C:\Windows\Temp\F343...	dismhost.exe (86A6A5C...	msense.exe	MsSense.exe	9fd99403b433857f23...
☐	> Aug 11, 2026 8:51...	dc01.corp.local	administrator	DismHost.exe	C:\Users\Administrator\...	dismhost.exe (7300081C...	cleanmgr.exe	cleanmgr.exe /autoclean...	9fd99403b433857f23...
☐	> Aug 11, 2026 3:42...	dc01.corp.local	system	DismHost.exe	C:\Windows\Temp\A9AE...	dismhost.exe (F158DFC1...	msense.exe	MsSense.exe	9fd99403b433857f23...
☐	> Aug 11, 2026 2:49...	dc01.corp.local	system	DismHost.exe	C:\Windows\Temp\7936...	dismhost.exe (1A866885...	msense.exe	MsSense.exe	9fd99403b433857f23...
☐	> Aug 11, 2026 2:49...	dc01.corp.local	system	DismHost.exe	C:\Windows\Temp\95B4...	dismhost.exe (CDE7749F...	msmpeng.exe	MsMpEng.exe	9fd99403b433857f23...
☐	> Aug 10, 2026 7:30...	win11-client01.corp...	socadmin	OneDriveLauncher.exe	C:\Users\Socadmin.COR...	"OneDriveLauncher.exe"...	svchost.exe	svchost.exe -k netwscs...	1e1a15ea90f7651f81...
☐	> Aug 10, 2026 7:26...	dc01.corp.local	administrator	DismHost.exe	C:\Users\Administrator\...	dismhost.exe (5497BEE7...	cleanmgr.exe	cleanmgr.exe /autoclean...	9fd99403b433857f23...
☐	> Aug 10, 2026 7:20...	win11-client01.corp...	socadmin	FileCoAuth.exe	C:\Users\Socadmin.COR...	"FileCoAuth.exe" -Embe...	svchost.exe	svchost.exe -k DcomLau...	68ab9a797cd3e04cb...
☐	> Aug 10, 2026 7:20...	win11-client01.corp...	socadmin	OneDrive.exe	C:\Users\Socadmin.COR...	"OneDrive.exe" /backgr...	explorer.exe	Explorer.EXE	3986df145c2602ec3f...

Evidence — user-writable-path execution results (91 events), including *OneDriveSetup.exe*, *FileCoAuth.exe*, *OneDrive.exe*, and *DismHost.exe*.

Assessment: The path alone is insufficient for a verdict. Each candidate requires a pivot through file path → parent process → command line → user → device → hash → network activity before any conclusion is drawn.

Analyst Conclusion

Suspicious-location candidates identified; no confirmed malicious execution established.

7. Hunt 3 — LOLBin Parent → Child Relationship Analysis

Hypothesis: Unusual execution chains involving dual-use or administrative binaries may expose suspicious execution patterns that a single-process view would miss.

Query

```

DeviceProcessEvents
| where FileName in~ (
  "powershell.exe", "cmd.exe", "wscript.exe", "cscript.exe",
  "mshta.exe", "rundll32.exe", "regsvr32.exe", "certutil.exe"
)
| summarize
  ExecutionCount = count()
  by InitiatingProcessFileName, FileName
| order by ExecutionCount asc

```

Result: 17 grouped relationships.

The screenshot shows a SIEM query interface. At the top, there's a 'Query' tab with a 'Run query' button. Below it, a KQL query is entered:

```

1 DeviceProcessEvents
2 | where FileName in- (
3   "powershell.exe",
4   "cmd.exe",
5   "mscript.exe",
6   "cscript.exe",
7   "mshta.exe",
8   "rundll32.exe",
9   "regsvr32.exe",
10  "certutil.exe"
11 )
12 | summarize
13   ExecutionCount = count()
14   by InitiatingProcessFileName, FileName
15 | order by ExecutionCount asc

```

Below the query, there's a 'Results' tab. It shows 17 items. The results are grouped by InitiatingProcessFileName and FileName, showing the ExecutionCount for each group. The results are sorted by ExecutionCount in ascending order.

InitiatingProcessFileName	FileName	ExecutionCount
cmd.exe	cmd.exe	1
control.exe	rundll32.exe	1
powershell.exe	powershell.exe	2
cmd.exe	powershell.exe	2
mpcmdrun.exe	powershell.exe	3
explorer.exe	rundll32.exe	4
cmd.exe	powershell.exe	10
explorer.exe	powershell.exe	12
mpcmdrun.exe	cmd.exe	15
compattelrunner.exe	powershell.exe	20
explorer.exe	cmd.exe	25
splunkd.exe	splunk-powershell.exe	34
cmd.exe	cmd.exe	34

Evidence — parent/child relationship results (17 groupings), including *windowsterminal.exe* → *cmd.exe*, *control.exe* → *rundll32.exe*, and *mpcmdrun.exe* → *powershell.exe*.

Assessment: These relationships are useful hunting leads, but parent/child rarity alone does not establish maliciousness. Required pivots include command line, account, device, timestamp, frequency, and any associated network activity.

Analyst Conclusion

Unusual execution relationships identified; additional contextual evidence would be required for a malicious verdict.

8. Hunt 4 — Single-Device Remote Destination Analysis

Hypothesis: Remote destinations contacted by only one device may identify infrastructure requiring further investigation.

Query

```

DeviceNetworkEvents
| where isnotempty(RemoteIP)
| summarize
    ConnectionCount = count(),
    Devices = dcount(DeviceName),
    Processes = dcount(InitiatingProcessFileName)
    by RemoteIP
| where Devices == 1
| order by ConnectionCount asc

```

Result: 605 one-device remote-IP groupings.

```

1 DeviceNetworkEvents
2 | where isnotempty(RemoteIP)
3 | summarize
4   ConnectionCount = count(),
5   Devices = dcount(DeviceName),
6   Processes = dcount(InitiatingProcessFileName)
7   by RemoteIP
8 | where Devices == 1
9 | order by ConnectionCount asc

```

RemoteIP	ConnectionCount	Devices	Processes
> 2600:1417:73:1...	1	1	1
> 2600:140f3:1a3...	1	1	1
> 2603:1046:2000...	1	1	1
> 2603:1046:2000...	1	1	1
> 2603:192.168.5.1...	1	1	1
> 2600:140f3:68...	1	1	1
> 2603:1046:2000...	1	1	1
> 2600:1417:73:1...	1	1	1
> 2600:140f6:17...	1	1	1
> 2603:1063:101...	1	1	1
> 2603:192.168.175...	1	1	1
> 2603:1046:2000...	1	1	1
> 2603:1046:2000...	1	1	1
> 2600:1417:73:1...	1	1	1
> 2603:1046:2000...	1	1	1
> 2603:192.168.22...	1	1	1

Evidence — single-device remote destination results (605 groupings), dominated by IPv6 and private/link-local addresses.

Assessment: This is a broad triage query, not a malicious-IP detector. Private, link-local, and IPv6 addresses can represent completely legitimate system/network activity. The captured evidence did not establish a malicious destination, and no external reputation check was performed against these addresses.

Analyst Conclusion

Network hunting candidates identified; no confirmed malicious destination established.

9. Hunt 5 — Authentication Pattern Analysis

Hypothesis: Unusual or low-frequency authentication patterns can identify accounts or logon types requiring further examination.

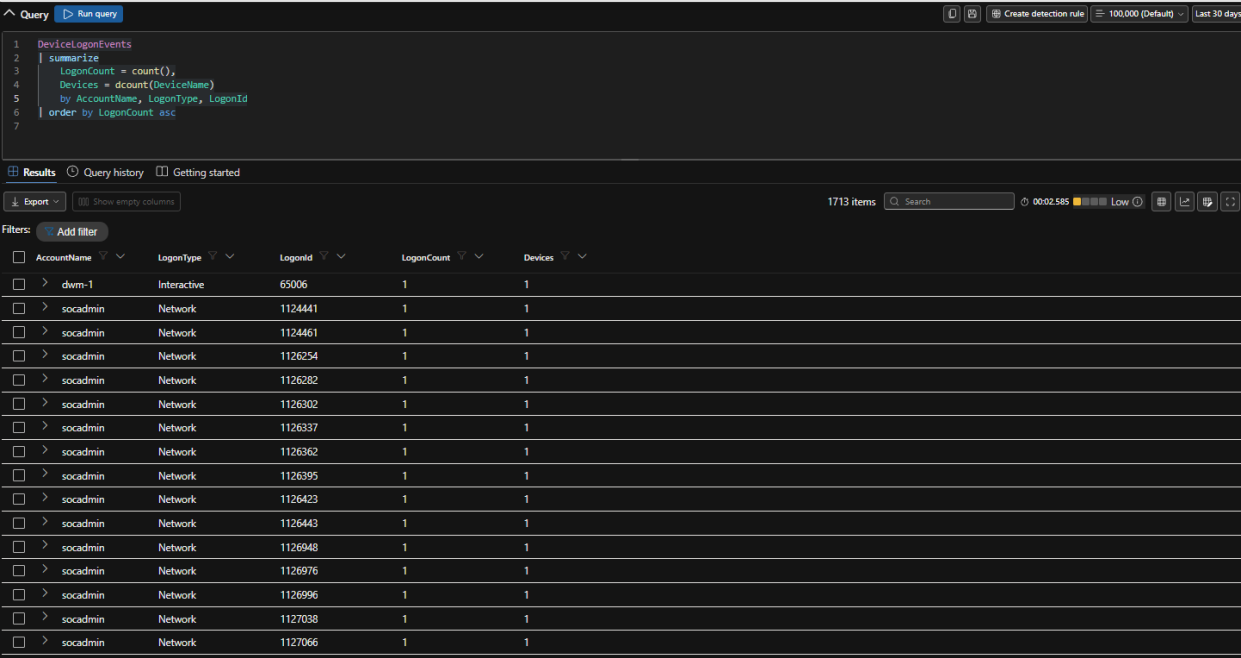
Query

```

DeviceLogonEvents
| summarize
  LogonCount = count(),
  Devices = dcount(DeviceName)
  by AccountName, LogonType, LogonId
| order by LogonCount asc

```

Result: 1,713 grouped rows.



Evidence — authentication pattern results (1,713 grouped rows), including one-off Interactive activity for dwm-1 and numerous Network-type entries for socadmin.

Assessment: Low-frequency authentication activity does not automatically indicate credential compromise. Additional context would be required, including source/destination device, account role, logon type, timestamp, surrounding process activity, authentication failures, and network context. This figure is a grouped query result, not a raw count of every logon event in the window.

Analyst Conclusion

Authentication baseline established; no confirmed account compromise identified from this hunt.

10. Supporting Hunts

The five primary hunts were supplemented with four additional queries that broadened coverage into persistence, LOLBin abuse, and PowerShell-specific behavior. These are documented more briefly — each still follows the same query → result → assessment discipline.

Supporting Hunt	Result	Purpose
Run / RunOnce registry	26	Persistence-context hunting
LOLBin execution	99	Dual-use binary hunting
Focused suspicious PowerShell	10	High-risk PowerShell keyword hunting
Broad PowerShell baseline	257	PowerShell activity baseline

10.1 Run / RunOnce Registry Hunting

Result: 26 events. The query targeted Software\Microsoft\Windows\CurrentVersion\Run and \RunOnce.

Query: `DeviceRegistryEvents`
`where RegistryKey has_any (`
 `@"Software\Microsoft\Windows\CurrentVersion\Run",`
 `@"Software\Microsoft\Windows\CurrentVersion\RunOnce"`
`)`
`project Timestamp,`
 `DeviceName,`
 `ActionType,`
 `RegistryKey,`
 `RegistryValueName,`
 `RegistryValueData,`
 `InitiatingProcessFileName,`
 `InitiatingProcessCommandLine`
`order by Timestamp desc`

Results: 26 items

Timestamp	DeviceName	ActionType	RegistryKey	RegistryValueName	RegistryValueData	InitiatingProcessFileName	InitiatingProcessCommandLine
Aug 12, 2026 9:01:...	win11-client01.corp.l...	RegistryValueSet	HKEY_CURRENT_USER\S...	Delete Cached Standalo...	C:\WINDOWS\system32...	onedrivesetup.exe	OneDriveSetup.exe /upd...
Aug 12, 2026 9:01:...	win11-client01.corp.l...	RegistryValueSet	HKEY_CURRENT_USER\S...	Delete Cached Update B...	C:\WINDOWS\system32...	onedrivesetup.exe	OneDriveSetup.exe /upd...
Aug 9, 2026 7:23:5...	win11-client01.corp.l...	RegistryValueSet	HKEY_CURRENT_USER\S...	MicrosoftEdgeAutoLaun...	"C:\Program Files (x86)...	msedge.exe	"msedge.exe" --no-start...
Aug 9, 2026 7:23:2...	win11-client01.corp.l...	RegistryValueSet	HKEY_CURRENT_USER\S...	MicrosoftEdgeAutoLaun...	"C:\Program Files (x86)...	msedge.exe	"msedge.exe" --single-a...
Aug 9, 2026 7:19:2...	win11-client01.corp.l...	RegistryValueSet	HKEY_CURRENT_USER\S...	OneDrive	"C:\Users\Socadmin.CO...	onedrive.exe	setautostart /background
Aug 9, 2026 7:10:1...	win11-client01.corp.l...	RegistryValueSet	HKEY_CURRENT_USER\S...	MicrosoftEdgeAutoLaun...	"C:\Program Files (x86)...	msedge.exe	"msedge.exe" --no-start...
Aug 8, 2026 8:15:4...	dc01.corp.local	RegistryValueSet	HKEY_LOCAL_MACHINE...	GrpConv	grpconv -o	vboxdrvinst.exe	"VBoxDrvinst.exe" --logf...

Evidence — Run/RunOnce persistence-context results (26 events), including OneDrive, Microsoft Edge auto-launch, and VirtualBox-related entries.

Registry Run/RunOnce locations are persistence-relevant, but legitimate applications commonly use them. No confirmed malicious persistence was established from these results.

10.2 LOLBin Execution Hunting

Result: 99 events across certutil.exe, mshta.exe, rundll32.exe, regsvr32.exe, wscript.exe, msixexec.exe, and bitsadmin.exe.

Query: `DeviceProcessEvents`
`where FileName in (`
 `"certutil.exe",`
 `"mshta.exe",`
 `"rundll32.exe",`
 `"regsvr32.exe",`
 `"wscript.exe",`
 `"msixexec.exe",`
 `"bitsadmin.exe"`
`)`
`project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName, InitiatingProcessCommandLine`
`order by Timestamp desc`

Results: 99 items

Timeline visualization: Shows a peak in events around August 9th.

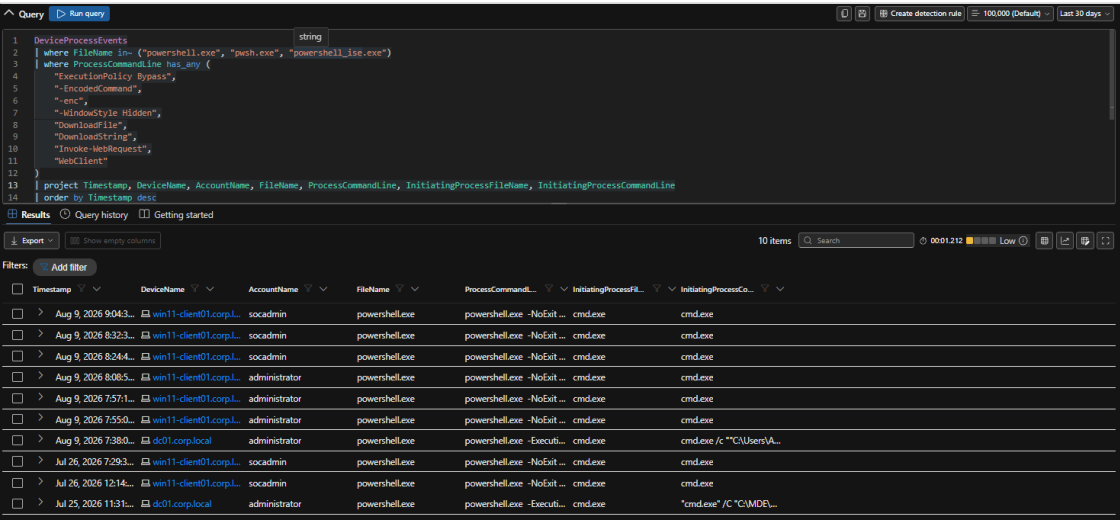
Timestamp	DeviceName	AccountName	FileName	ProcessCommandLine	InitiatingProcessFileName	InitiatingProcessCommandLine
Aug 12, 2026 8:26:...	win11-client01.corp.l...	socadmin	rundll32.exe	rundll32.exe C:\WINDO...	svchost.exe	svchost.exe -k wsappx ...
Aug 12, 2026 7:56:...	win11-client01.corp.l...	system	rundll32.exe	"rundll32.exe" C:\WIND...	svchost.exe	svchost.exe -k netwics ...
Aug 12, 2026 7:55:...	dc01.corp.local	system	rundll32.exe	rundll32.exe C:\Windo...	svchost.exe	svchost.exe -k netwics ...
Aug 12, 2026 7:55:...	win11-client01.corp.l...	system	rundll32.exe	"rundll32.exe" /d acpro...	svchost.exe	svchost.exe -k netwics ...
Aug 12, 2026 12:3...	dc01.corp.local	system	rundll32.exe	rundll32.exe /d acpro...	svchost.exe	svchost.exe -k netwics ...
Aug 12, 2026 11:1...	win11-client01.corp.l...	system	rundll32.exe	"rundll32.exe" /d acpro...	svchost.exe	svchost.exe -k netwics ...

Evidence — LOLBin execution results (99 events), predominantly rundll32.exe invoked via svchost.exe.

The presence of a LOLBin is not itself malicious. Each candidate must be examined alongside command line, parent process, user, file path, and network activity to distinguish legitimate administrative or software activity from potential abuse.

10.3 Focused PowerShell Hunting

Result: 10 events, filtered on keywords including ExecutionPolicy Bypass, -EncodedCommand, WindowStyle Hidden, DownloadFile/DownloadString, and Invoke-WebRequest/WebClient.

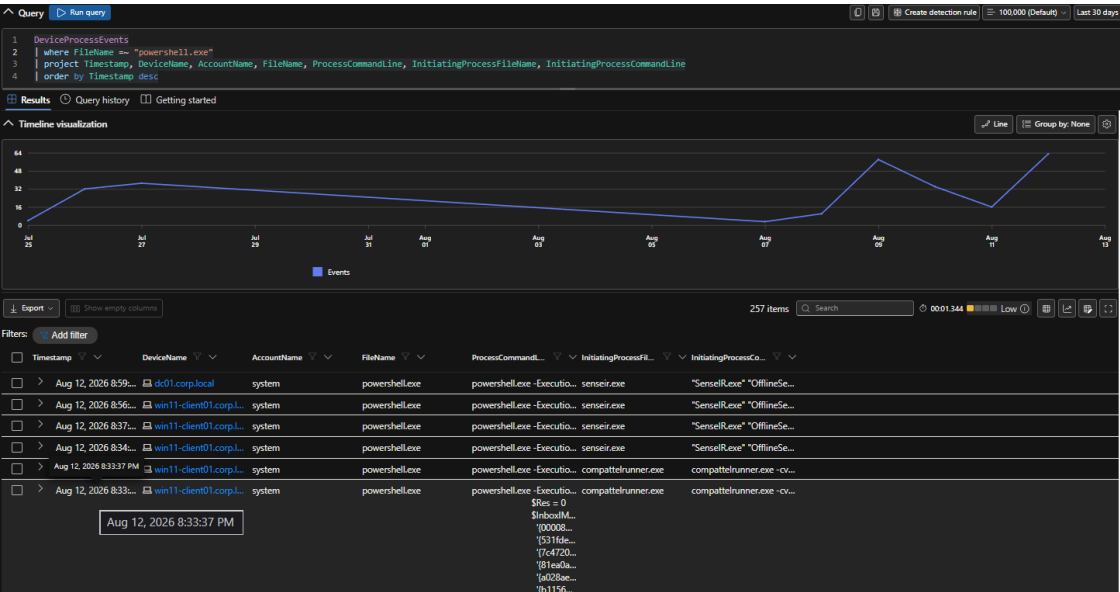


Evidence — focused suspicious PowerShell results (10 events), all launched via cmd.exe under socadmin/administrator.

These terms are useful hunting indicators because they can occur in suspicious PowerShell execution, but keyword matching is not proof of malicious activity — administration, software deployment, and security tooling can legitimately generate similar telemetry.

10.4 Broad PowerShell Baseline

Result: 257 events for FileName == "powershell.exe" with no keyword filter.



Evidence — broad PowerShell baseline (257 events), with SenseIR.exe and compattelrunner.exe visible as initiating processes.

This baseline demonstrates why a hunting analyst establishes a volume baseline before interpreting suspicious-looking PowerShell: if PowerShell is common in the environment, simply finding PowerShell is a weak signal on its own — the investigation must move toward command line, parent, user, device, time, and network context.

11. Cross-Hunt Analysis

The strongest part of threat hunting is not the individual query — it is the pivot between telemetry sources. A suspicious PowerShell event, for example, can be investigated by moving from DeviceProcessEvents into DeviceNetworkEvents, DeviceRegistryEvents, and DeviceLogonEvents in turn. Likewise, a suspicious registry modification can be pivoted back to the process responsible for making it.

Rare process

```

|
Process command line
|
Parent process
|
User
|
Registry activity
|
Network connections
|
Authentication
|
Final assessment

```

No single entity in this project's results appeared across multiple hunts in a way that would justify escalation — that absence is itself a data point, and is recorded honestly in Section 12 rather than manufactured.

12. Findings

Finding	Status	Evidence-Based Basis
Process telemetry available	Confirmed	11,229 process events observed in the baseline query
Registry telemetry available	Confirmed	9,376 registry events observed
Network telemetry available	Confirmed	8,216 network events observed
Rare process candidates	Requires triage	77 low-frequency candidates identified
User-writable execution	Requires triage	91 matching events identified
LOLBin parent/child relationships	Requires triage	17 low-frequency relationships identified
Single-device remote destinations	Requires triage	605 one-device remote-IP groupings identified
Authentication pattern groupings	Established	1,713 grouped results returned
Confirmed malicious process	Not established	Captured evidence insufficient
Confirmed malicious persistence	Not established	Captured evidence insufficient
Confirmed C2 destination	Not established	Captured evidence insufficient
Confirmed account compromise	Not established	Captured evidence insufficient
Confirmed compromise (overall)	Not established	No captured hunt independently established compromise

13. Analyst Lessons

- Rarity is not maliciousness. Rare software can be legitimate, newly installed, administrative, security-related, or maintenance-related.
- Context matters more than a single field. A suspicious path becomes meaningful only combined with a suspicious parent, command line, account, timing, or network connection.
- Baselines reduce false positives. The broad PowerShell hunt showed PowerShell was common in this environment, so powershell.exe alone is a weak hunting signal here.
- Hunting is iterative: query → interesting result → pivot → new query → context → assessment, rarely resolved on the first pass.
- Analysts must be comfortable saying "not enough evidence." The correct SOC conclusion is not "unusual, therefore malicious" — it is "warrants further investigation, but available evidence is insufficient to establish maliciousness."

14. Limitations

- The portal was consistently operated using the Last 30 days window; Microsoft's standard Advanced Hunting experience documents up to 30 days of raw Defender data.
- Screenshots represent the captured query results and are not a complete export of every underlying event.
- Rarity-based queries naturally surface legitimate software and infrastructure activity alongside genuine anomalies.
- The Hunt 4 remote-IP results included private, link-local, and IPv6 addresses; these were not treated as malicious without supporting evidence, and no external reputation check was performed against them.
- No candidate from any hunt was promoted to a confirmed incident without corroborating evidence.
- This was a controlled SOC-lab assessment, not a production-wide threat assessment.
- The Hunt 5 authentication figure (1,713) represents grouped query results, not a raw count of all authentication events in the window.

15. Project Outcome

Capability	Demonstrated
KQL fundamentals	Yes
Advanced Hunting	Yes
Hypothesis-driven hunting	Yes
Process, persistence, LOLBin, PowerShell, network, and authentication analysis	Yes
Baseline construction and rarity analysis	Yes
Cross-table pivoting methodology	Yes
False-positive / benign-positive reasoning	Yes
Evidence-based, conservative conclusions	Yes
Confirmed compromise	Not established

16. Conclusion

Project 3 demonstrates a complete proactive threat-hunting workflow using Microsoft Defender XDR Advanced Hunting and KQL. Rather than waiting for an alert, the investigation began with hypotheses and used endpoint, registry, network, and authentication telemetry to identify unusual activity — surfacing rare processes, user-writable-path execution, unusual process relationships, uncommon network destinations, and low-frequency authentication patterns as candidates for further investigation.

None of the captured evidence independently established a confirmed compromise, and that conclusion is intentional: a competent threat hunter must be able to distinguish unusual ≠ malicious ≠ confirmed compromise. This project demonstrates analytical discipline and evidence-based decision-making, not just KQL syntax.

Together with the earlier portfolio projects, this completes three distinct stages of SOC analyst workflow: Project 1 (Endpoint Investigation, Response & Automated Disruption) is reactive investigation and response; Project 2 (Threat Intelligence & IOC Management) is IOC operationalization; Project 3 (Advanced Hunting & Proactive Threat Hunting) is proactive discovery and hypothesis-driven analysis — three different competencies, not three versions of the same investigation.

17. Evidence Register

Evidence	Purpose
DeviceProcessEvents baseline screenshot	Confirms process telemetry availability (11,229 events)
DeviceRegistryEvents baseline screenshot	Confirms registry telemetry availability (9,376 events)
DeviceNetworkEvents baseline screenshot	Confirms network telemetry availability (8,216 events)
Hunt 1 — rare process screenshot	Validates rare-process query (77 candidates)
Hunt 2 — user-writable path screenshot	Validates user-writable-path query (91 events)
Hunt 3 — parent/child screenshot	Validates parent/child relationship query (17 groupings)
Hunt 4 — single-device remote destination screenshot	Validates single-device remote-destination query (605 groupings)
Hunt 5 — authentication pattern screenshot	Validates authentication pattern query (1,713 grouped rows)
Run/RunOnce registry screenshot	Validates persistence-context query (26 events)
LOLBin execution screenshot	Validates LOLBin query (99 events)
Focused PowerShell screenshot	Validates focused keyword query (10 events)
Broad PowerShell baseline screenshot	Validates PowerShell volume baseline (257 events)
Project_3_Defender_XDR_KQL_Hunting_Queries.kql	All 12 queries used in this project, in execution order

18. Official References

#	Source
1	Advanced hunting overview — learn.microsoft.com/en-us/defender-xdr/advanced-hunting-overview
2	Advanced hunting query language — learn.microsoft.com/en-us/defender-xdr/advanced-hunting-query-language
3	Advanced hunting schema tables — learn.microsoft.com/en-us/defender-xdr/advanced-hunting-schema-tables
4	DeviceRegistryEvents table — learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceregistryevents-table
5	DeviceNetworkEvents table — learn.microsoft.com/en-us/defender-xdr/advanced-hunting-devicenetworkevents-table
6	DeviceLogonEvents table — learn.microsoft.com/en-us/defender-xdr/advanced-hunting-devicelogonevents-table

19. Investigation Sign-Off

Field	Value
Prepared by	Edwin — SOC Analyst (L1)
Investigation type	Controlled enterprise-style SOC laboratory
Scope statement	Controlled SOC lab portfolio artifact; no production incident is implied

Field	Value
Disposition	Complete — hunting leads documented; no confirmed compromise established